# F005 v0.5.0 - INDEPENDENT LINEARIZED EULER SLOSHING

Evidence-first / no fitted frequency / no damping fit / not nonlinear Euler

**Status: DEMO_SYNTHETIC_NOT_EXPERIMENT**

## Source and model boundary

Potential flow linearized around still water, rectangular 2D tank, impermeable rigid walls/bottom, small free-surface displacement.

k_n = (2n+1)pi/L; omega_n^2 = (g*k_n + sigma*k_n^3/rho) * tanh(k_n*h)

q_n" + omega_n^2*q_n = -k_n*tanh(k_n*h)*b_n*a(t)

b_n=4(-1)^n/(L k_n^2), eta(x,t)=sum_n q_n(t)sin(k_n x).

Forcing uses publisher acceleration. Zero damping is enforced as inviscid Euler; friction/contact/meniscus/3D flow remain unmodelled.

## Gates

E0_GEOMETRY: DEMO_ONLY

E1_MODAL_SOLVER: PASS_SYNTHETIC

E2_INPUT_INTEGRITY: SYNTHETIC

E3_FROM_REST: NOT_RUN

E4_CONDITIONAL_ONE_MODE: NOT_RUN

E5_BASELINE_COMPARISON: NOT_RUN

E6_FAMILY_DISJOINT: DEVELOPMENT_PROXY_ONLY

E7_PKLSA: NOT_RUN

E8_SST_CLAIMS: UNRESOLVED

E9_LINEARIZATION: NOT_RUN

## Inputs and provenance

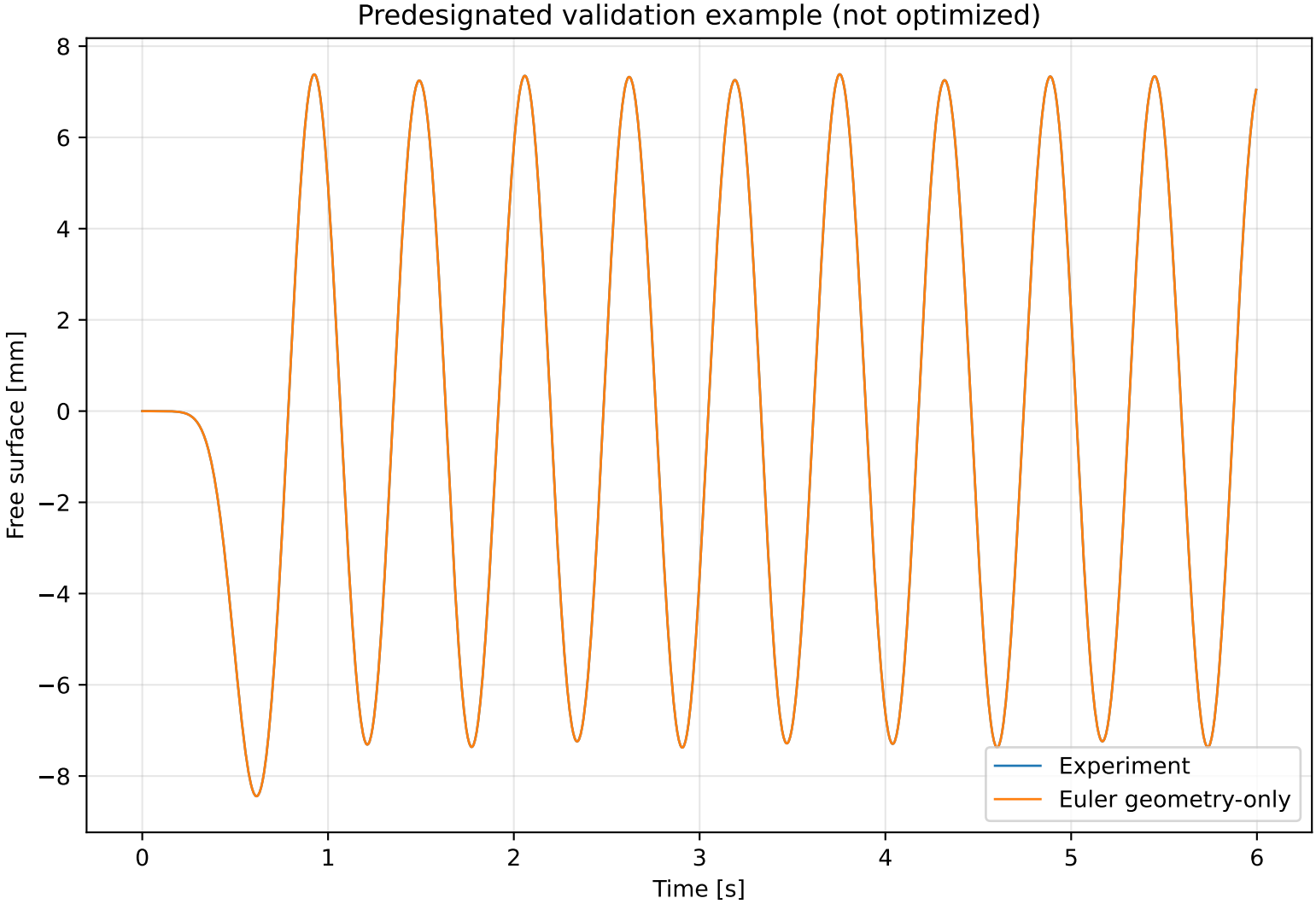
- 0.4.3 unchanged; frozen TEST/OOD response signals not indexed.

- Geometry cannot be inferred from T75 designation or prior ARX/oscillator fitted frequency.

- Without verified dimensions and sensor polarity, no physical forecast is reported.

- ALL values are deliberately hypothetical; this output cannot validate the Zenodo experiment.

**Do NOT promote: physical SST and source-native PKLSA remain UNRESOLVED.**


Predesignated validation example (not optimized)
Free surface [mm]
Time [s]
Experiment
Euler geometry-only